

CSA1703 – Artificial Intelligence

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Slot – A

2. 8 QUEEN PROBLEM

Code:

```
print("B.Deepthi reddy")
print(192312288)
print("Queen Problem")
N = int(input("Enter the number of queens: "))
x = [-1] * N
def safe(r, c):
    for i in range(c):
        if x[i] == r or abs(x[i] - r) == abs(i - c):
            return False
    return True
def queen(c):
    if c == N:
        print("\nSolution:")
        for i in range(N):
            print(x[i], end=" ")
        return True
```

```

for r in range(N):
    if safe(r, c):
        x[c] = r
        if queen(c + 1):
            return True
return False

```

```

if not queen(0):
    print("No Solution")

```

Output:

main.py	Run	Output
<pre> 1 2 print("B.Deepthi reddy") 3 print(192312288) 4 print("Queen Problem") 5 6 7 N = int(input("Enter the number of queens: ")) 8 9 x = [-1] * N 10 11 def safe(r, c): 12 for i in range(c): 13 if x[i] == r or abs(x[i] - r) == abs(i - c): 14 return False 15 return True 16 17 def queen(c): </pre>	Run	<pre> B.Deepthi reddy 192312288 Queen Problem Enter the number of queens: 6 Solution: 1 3 5 0 2 4 === Code Execution Successful === </pre>